



# Fundamentals of Information Theory

## Asymptotic Equipartition Property

渐进均分性

Yayu Gao

School of Electronic Information and Communications  
Huazhong University of Science and Technology

Email: [yayugao@hust.edu.cn](mailto:yayugao@hust.edu.cn)

# Outline

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- Motivation
- Law of large numbers
- Asymptotic Equipartition Property (AEP)
- Typical Set
- Insights of the AEP

**AEP is central in information theory.**



# 本节学习目标

1. 理解效率与可靠性之间的折衷关系
2. 写出AEP定理
3. 写出AEP的证明过程
4. 概述AEP与大数定律之间的关系
5. 写出典型集的定义
6. 说出AEP对信源编码的物理意义

## 重难点:

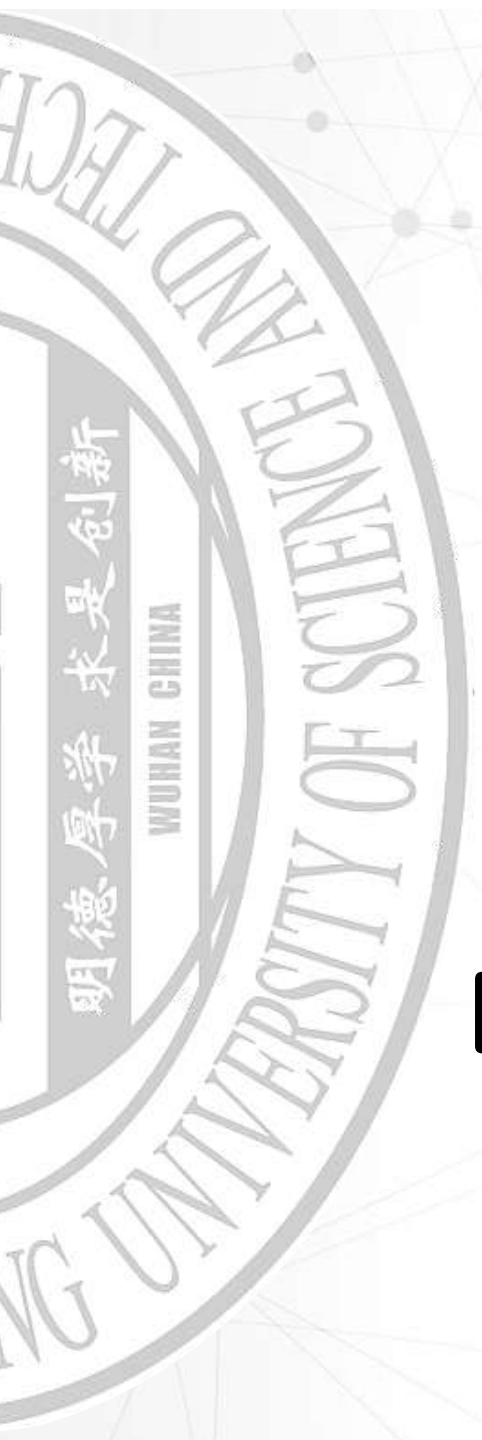
- AEP定理及证明
- AEP的物理意义

# 01

## Motivation

### Efficiency vs. Reliability

**Core Problem of Communication Systems**



# **Major challenges** for establishing a communication theory

- How much information is transmitted?
  - **Quantify** information carried by symbols
- How to evaluate the performance of communication systems?
  - Transmission **efficiency** in communication systems
  - **Accuracy** of information transmission
- Nature of real-world scenarios?
  - Noise interference
- Core problems: **efficiency vs. reliability**
- Pioneering work by Claude E. Shannon and Norbert Wiener



# Efficiency vs. Reliability



**"You Must Have Spent  
Years on Shorthand"**

***"No! I Learned in 6 WEEKS!"***

**S**PEEDWRITING, the ABC shorthand, can be completely mastered in one-fourth the time required by symbol systems and is far easier and more accurate to write and transcribe. Tens of thousands of shorthand writers have been freed from the drudgery of old-fashioned methods of learning and writing shorthand by the marvelous SPEEDWRITING System. It has no signs, no symbols, no machines, but is built on the familiar letters of the alphabet—the ABC's you already know!

**Qualify as a Fast, Accurate Shorthand  
Writer in 72 Hours of Home Study**

***Speedwriting***

*f u cn rd ths, u cn  
bcm a sec & gt  
a gd jb w hi pa*

- f u cn rd ths, u cn bcm a sec & gt a gd jb w hi pa!
- If you can read this, you can become a secretary and get a good job with high pay!

# Efficiency vs. Reliability

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- The eternal issues of information theory
  - Lose reliability to achieve higher efficiency
  - Lose efficiency to achieve higher reliability
- Balance between efficiency and reliability
  - **Efficiency**
    - digital case : send as few symbols as possible
    - analog case : reduce the time that the channel or the bandwidth is used
  - **Reliability**
    - digital case : reduce the error probability as small as possible
    - analog case : reduce the noise as much as possible



# Source coding vs. Channel coding

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- **Source Coding**

- Core problem: **efficiency**
- Efficiency: having an average code length that is as small as possible
- Example: to use shorter code for the English letters which appear frequently, so as to reduce the average code length

- **Channel Coding**

- Core problem: **reliability**
- Reliability: to cope with the errors in the transmission
- Example: to send the same sequence multiple times, so as to recover from the errors in channel





# Efficiency vs. Reliability: Redundancy

- 信源有冗余，可进行压缩
- 信源编码：
- 信源编码是通过尽可能压缩信源冗余度的手段，实现提高通信有效性的目的。

例：中华人民共和国      压缩 →      中国      效率最高

显然，压缩后信源的冗余度越低，通信的**有效性**越好。但另一方面，信源冗余度过低，甚至没有冗余度，又会带来通信**可靠性**方面的问题。

# Efficiency vs. Reliability: Redundancy

若通信过程中出现错误：

## 1. 当信源无冗余度

中国 → × 国 美国 法国 德国...?

中国 → 中× 中国 中央 中间...?

## 2. 当信源存在一定冗余度

中华人民  
共和国 → × 华人民  
× 和国 恢复 → 中华人民  
共和国

**结论：**通信有效性（信源编码）与可靠性（信道编码）

往往是一对矛盾。



# Efficiency vs. Reliability: Redundancy

**Q: How to represent the information of the source efficiently?**

- ✓ When is the entropy of a discrete random variable **maximized**?
- ✓ **How** to approach this information bound?

Let us begin with an example.

# Example #1: entropy of uniform *r.v.*

- Consider a uniform random variable with  $M = 8 = 2^3$  possible outcomes
- The bit strings of length  $\log_2 M = 3$  are sufficient to describe each outcome
- The entropy of this *r.v.* is

$$H(X) = - \sum_{i=1}^8 p(x) \log_2[p(x)] = \sum_{i=1}^8 \frac{1}{8} \log_2 8 = 3 \text{ bits}$$

- $H(X) = \log_2 M$

$x_1 \rightarrow 000$   
 $x_2 \rightarrow 001$   
 $x_3 \rightarrow 010$   
 $x_4 \rightarrow 011$   
 $x_5 \rightarrow 100$   
 $x_6 \rightarrow 101$   
 $x_7 \rightarrow 110$   
 $x_8 \rightarrow 111$

## Example #2: entropy of non-uniform r.v.

- Consider a **non-uniform** random variable with  $M = 8 = 2^3$  possible outcomes and probabilities  $(1/2, 1/4, 1/8, 1/16, 1/64, 1/64, 1/64, 1/64)$
- The bit strings of length  $\log_2 M = 3$  are sufficient to describe each outcome
- The entropy of this r.v. is

$$\begin{aligned} H(X) &= -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) - \frac{1}{8} \log_2\left(\frac{1}{8}\right) \\ &\quad - \frac{1}{16} \log_2\left(\frac{1}{16}\right) - \frac{4}{64} \log_2\left(\frac{1}{64}\right) \\ &= 2\text{bits} \end{aligned}$$

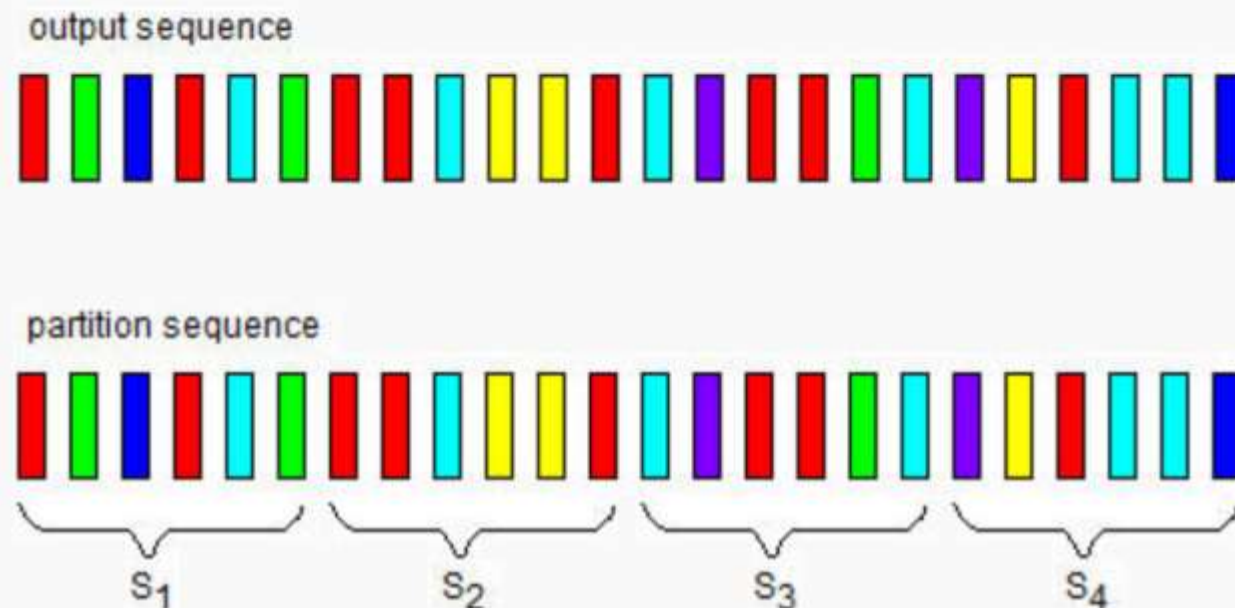
- Entropy and  $\log_2(M)$  disagree ( $H(X) < \log_2 M$ )
- The average length of source code word can be made shorter than 3

$x_1 \rightarrow 000$   
 $x_2 \rightarrow 001$   
 $x_3 \rightarrow 010$   
 $x_4 \rightarrow 011$   
 $x_5 \rightarrow 100$   
 $x_6 \rightarrow 101$   
 $x_7 \rightarrow 110$   
 $x_8 \rightarrow 111$



# Asymptotic Equipartition Property: Motivation

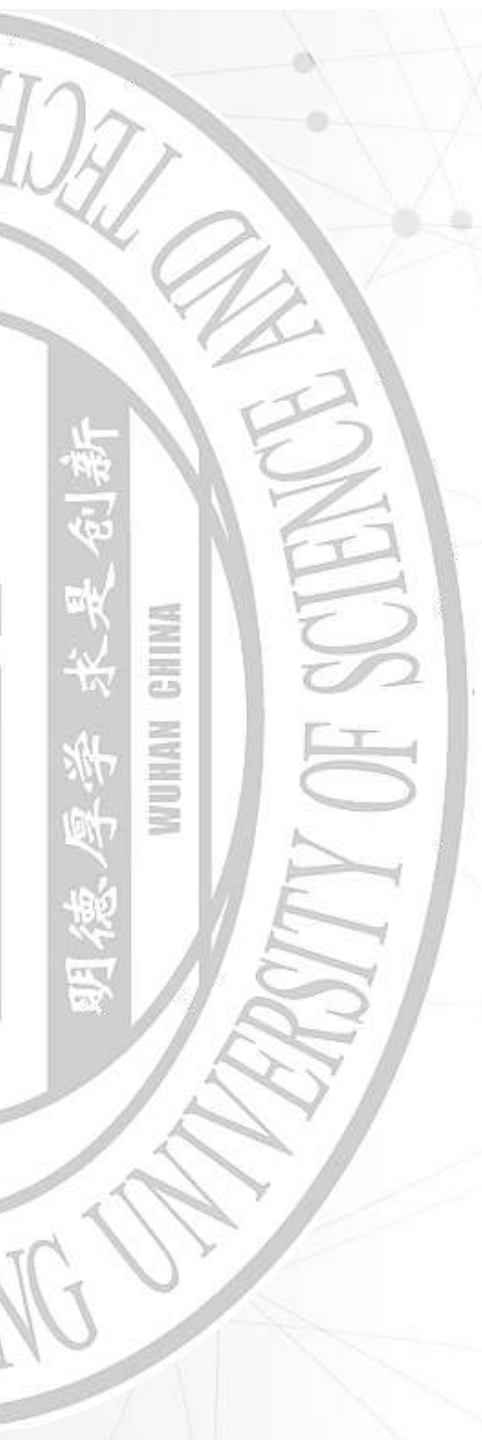
- Theorem: when the probability of a random variable is **uniformly** distributed, its discrete entropy is **maximized**.
- The output of an information source is usually **not** uniformly distributed.
- **Question:** How to transfer non-uniform distributed output into uniformly distributed output?





# 02

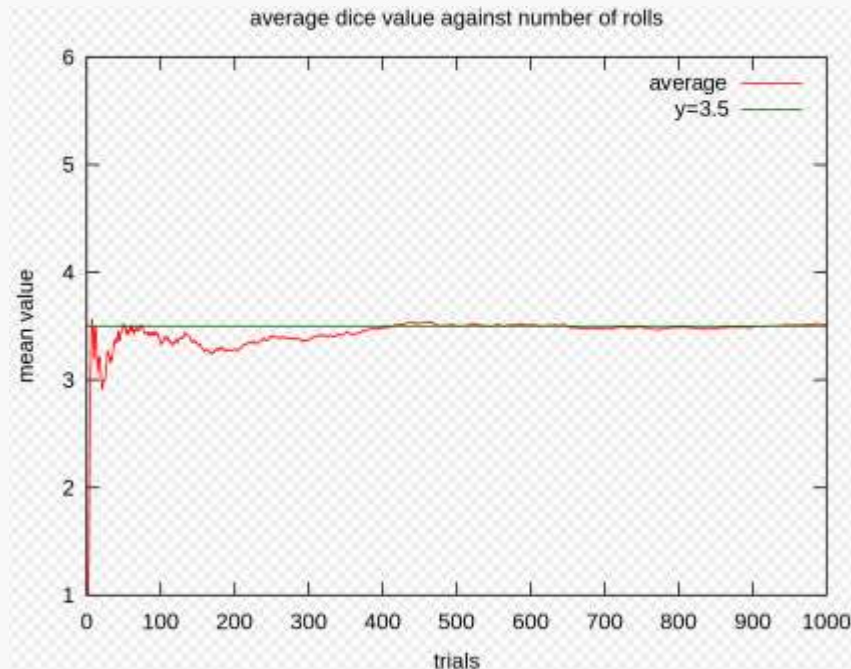
## Law of Large Number



# Asymptotic Equipartition Property: Motivation

- A single roll of a six-sided die produces one of the numbers 1, 2, 3, 4, 5, 6, each with equal probability.
- Therefore, the expected value of a single dice roll is

$$\frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5.$$



# Law of large numbers in classic probability theory

Bernoulli's law of large numbers

[http://en.wikipedia.org/wiki/Law\\_of\\_large\\_numbers](http://en.wikipedia.org/wiki/Law_of_large_numbers)



- In a Bernoulli experiment sequence with the probability  $p$ .

$$X_i = \begin{cases} 1, & A \text{ occurs at } i\text{-th experiment.} \\ 0, & A \text{ does not occur at } i\text{-th experiment.} \end{cases}$$

$$S_n = \sum_{i=1}^n X_i, n = 1, 2, \dots$$

- Weak law of large number:  $\frac{S_n}{n} \xrightarrow{P} p$
- A general law of large numbers is as follows:
- If  $X_i$  is i.i.d,

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} EX_i.$$

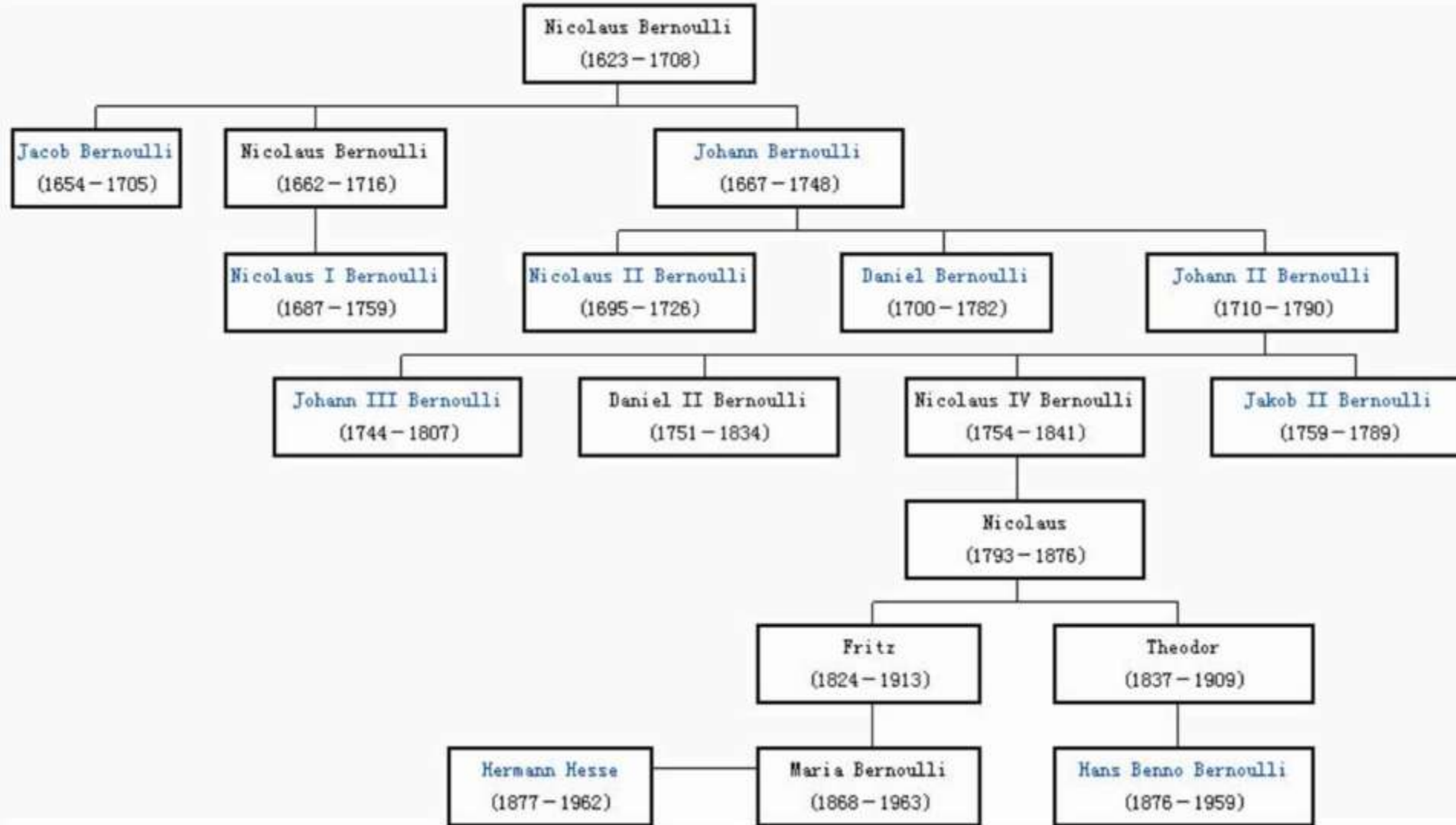
- Strong law of large number

$$P \left( \lim_{n \rightarrow \infty} |\mu_n - \mu| < \varepsilon \right) = 1$$

- Weak law of large number

$$\forall \varepsilon > 0, \lim_{n \rightarrow \infty} P (|\mu_n - \mu| < \varepsilon) = 1$$

# Bernoulli family tree





# Bernoulli family tree



- Jakob Bernoulli
- Probability Theory
- Bernoulli Law of Large Numbers



- Johan Bernoulli
- Calculus
- Teacher of Euler, L'Hospital...



- Daniel Bernoulli
- Math, Physics...
- Hydrodynamica

**"Invito patre sidera verso." 我违父意，钻研群星。**

# 03

## **Asymptotic Equipartition Property (AEP)**



# AEP property for *i.i.d. r.v.*

- We investigate the entropy characteristics of joint random variables from the simplest case.
- Independent and identically distributed (*i.i.d.*)
  - $X_i$  have same entropy  $H(X)$ ,

$$H(X_i) = - \sum p(X_i) \log p(X_i) = H(X).$$

- Weak law of large numbers

For *i.i.d.*  $X_i$ ,  $\frac{1}{n} \sum_{i=1}^n X_i$  is close to its expected value  $E(X_i)$  for large value of  $n$ .

# AEP property for *i.i.d.* r.v.: **theorem**

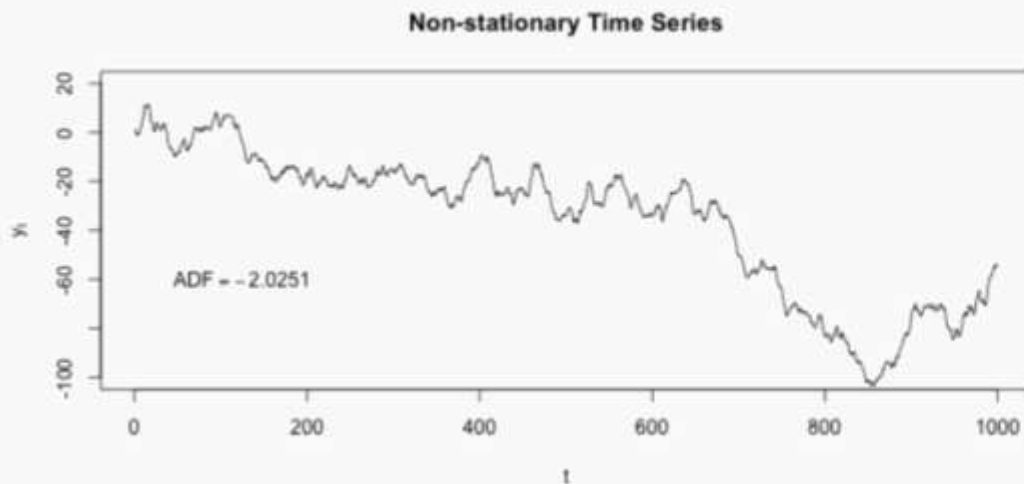
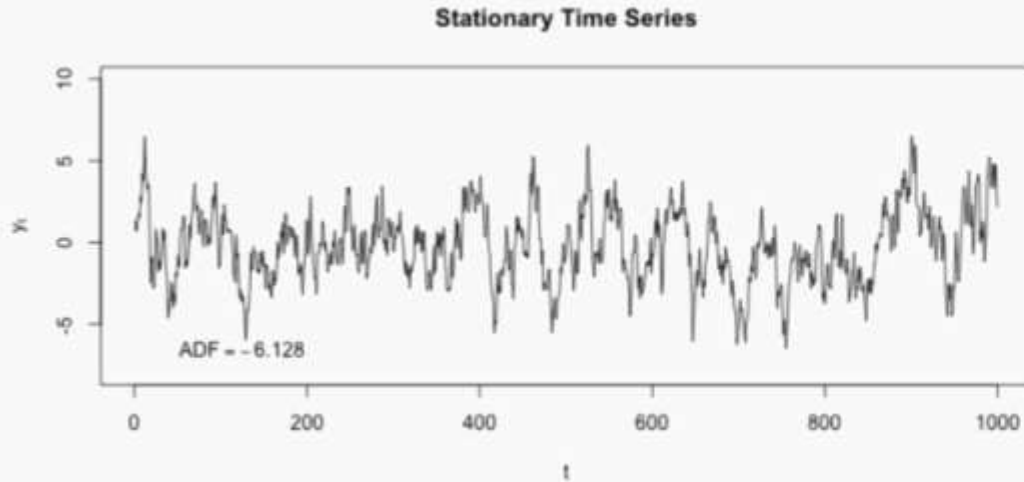
A direct consequence of the weak law of large numbers

If  $X_1, X_2, \dots, X_n$  are *i.i.d.*  $\sim p(x)$ , then  
 $-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X)$  in probability.

AEP was first stated by Shannon in his original 1948 paper, where he proved the results for *i.i.d.* processes and stated the result for **stationary ergodic processes**.

- What is stationary? What is ergodic?
- Can you prove the AEP?

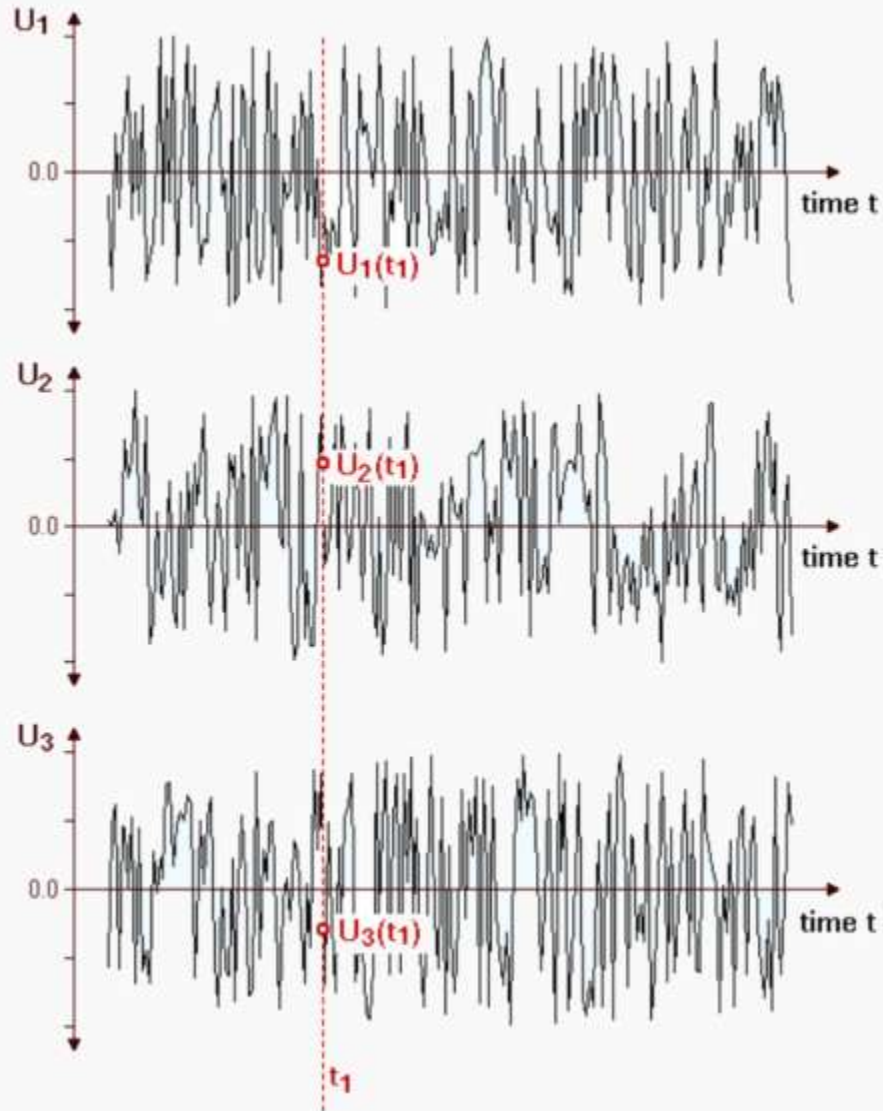
# Stationary processes



- A stochastic process  $\{X(t), t \geq 0\}$  is said to be a stationary process if for all  $n, s, t, \dots, t_n$ , the random vectors  $X(t_1), \dots, X(t_n)$  and  $X(t_1 + s), \dots, X(t_n + s)$  have the same joint distribution.
- In mathematics, a stationary process (or strict(ly) stationary process or strong(ly) stationary process) is a stochastic process whose **joint probability distribution does not change when shifted in time or space.**



# Ergodic processes



- In mathematics, the term ergodic is used to describe a dynamical system which, broadly speaking, has the same behavior averaged over time as averaged over the space of all the system's states (phase space).
- In statistics, the term describes a random process for which **the time average of one sequence of events is the same as the ensemble average**, and vice versa.

# AEP property for *i.i.d. r.v.*: proof

- Functions of independent random variables are also independent random variables.
- Thus, since the  $X_i$  are *i.i.d.*, so are  $\log [p(X_i)]$ .
- Hence, by the weak law of large numbers,

$$\begin{aligned} & -\frac{1}{n} \log [p(X_1, X_2, \dots, X_n)] \\ &= -\frac{1}{n} \log [p(X_1)p(X_2) \dots p(X_n)] \\ &= -\frac{1}{n} (\log (p(X_1)) + \log (p(X_2)) + \dots + \log (p(X_n))) \\ &= -\frac{1}{n} \sum_i \log p(X_i) \quad \rightarrow -E [\log p(X)] \text{ in probability} \\ & \quad \quad \quad = H(X) \end{aligned}$$

Weak law of large number

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} EX_i.$$



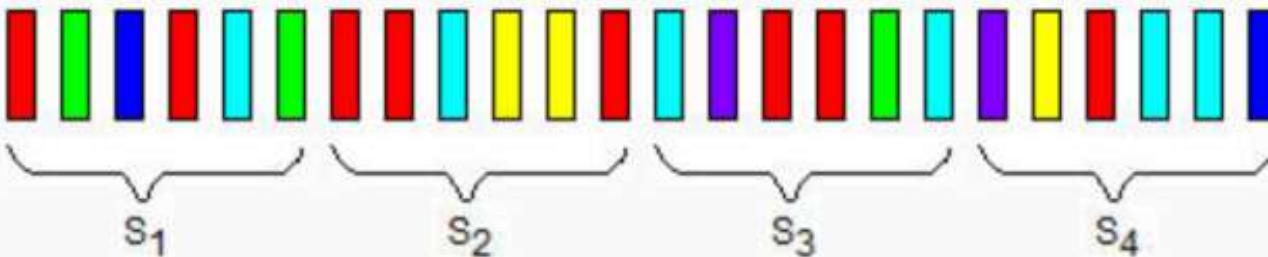
# AEP property for *i.i.d. r.v.*: comments

- The probability  $p(X_1, X_2, \dots, X_n)$  assigned to an observed sequence will be close to  $2^{-nH(X)}$ .
- Almost all events are almost equally surprising.
- How to represent an information source efficiently?

output sequence

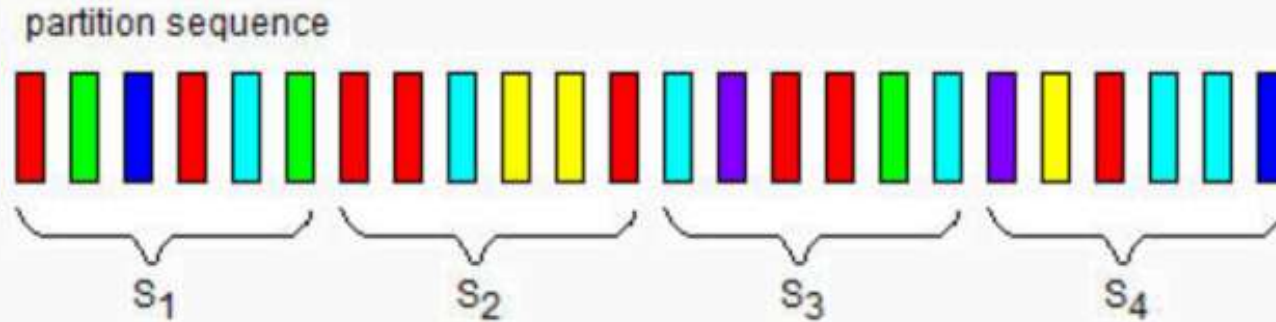


partition sequence



# AEP property for *i.i.d. r.v.*: **comments**

- Source coding

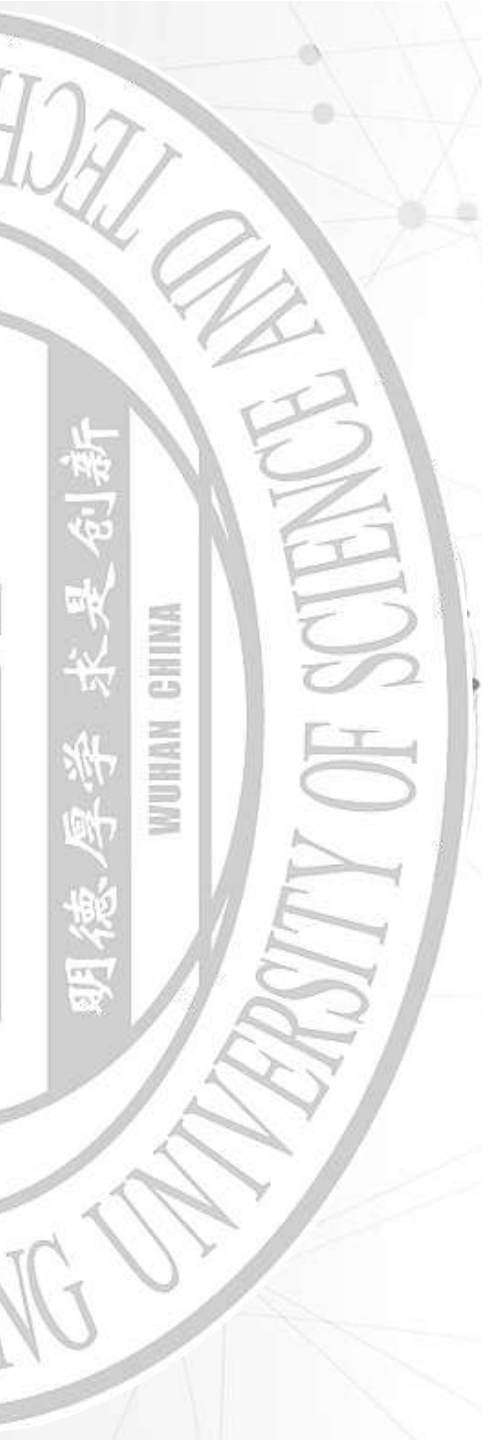


Partition sequence → Equal Probability → Maximum Entropy → **Fixed-length coding**

$nH(X)$  bits suffice on the average to describe  $n$  independent and *i.i.d. r.v.*'s.

# 04

## Typical Set



# What is Typical Set?

**Definition** The *typical set*  $A_\epsilon^{(n)}$  with respect to  $p(x)$  is the set of sequences  $(x_1, x_2, \dots, x_n) \in \mathcal{X}^n$  with the property

$$2^{-n(H(X)+\epsilon)} \leq p(x_1, x_2, \dots, x_n) \leq 2^{-n(H(X)-\epsilon)}.$$

- 所有典型序列的集合为**典型集** $A_\epsilon^{(n)}$ ，所有非典型序列的集合为**非典型集**。
- 通俗地讲，概率约等于 **$2^{-nH(X)}$** 的序列为典型序列。
- The typical set has probability **nearly 1**, all elements of the typical set are nearly equiprobable, and the number of elements in the typical set is nearly  **$2^{nH(X)}$** .

$$\mathbb{P}\{A_\epsilon^{(n)}\} \rightarrow 1, \text{ when } n \rightarrow \infty, \epsilon \rightarrow 0$$

$$|A_\epsilon^{(n)}| \approx 2^{nH(X)}, \text{ when } n \rightarrow \infty, \epsilon \rightarrow 0$$

# What is Typical Set?

1. If  $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$ , then  $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon$ .
2.  $\Pr\{A_\epsilon^{(n)}\} > 1 - \epsilon$  for  $n$  sufficiently large.
3.  $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$ , where  $|A|$  denotes the number of elements in the set  $A$ .
4.  $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X)-\epsilon)}$  for  $n$  sufficiently large.





# Example #1

- 例1:  $X_1, X_2, \dots, X_n$  为多符号离散无记忆信源, 每个符号服从均值为  $\frac{1}{2}$  的伯努利分布,

$$\mathbb{P}(X = 0) = \frac{1}{2}, \quad \mathbb{P}(X = 1) = \frac{1}{2}$$

判断  $x^n$  中哪些序列为典型序列。

- 解: 首先计算

$$H(X) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1$$

对于  $x^n$  中任一序列  $(x_1, x_2, \dots, x_n)$ , 如果其概率约等于

$$2^{-nH(X)} = 2^{-n}$$

则为典型序列。由此可见,  $x^n$  中所有序列都是典型的。典型序列的个数约为  $2^n$

## Example #2

- 例2:  $X_1, X_2, \dots, X_n$  为多符号离散无记忆信源, 每个符号服从分布,

$$\mathbb{P}(X = 0) = \frac{25}{32}, \quad \mathbb{P}(X = 1) = \mathbb{P}(X = 2) = \dots \mathbb{P}(X = 7) = \frac{1}{32}$$

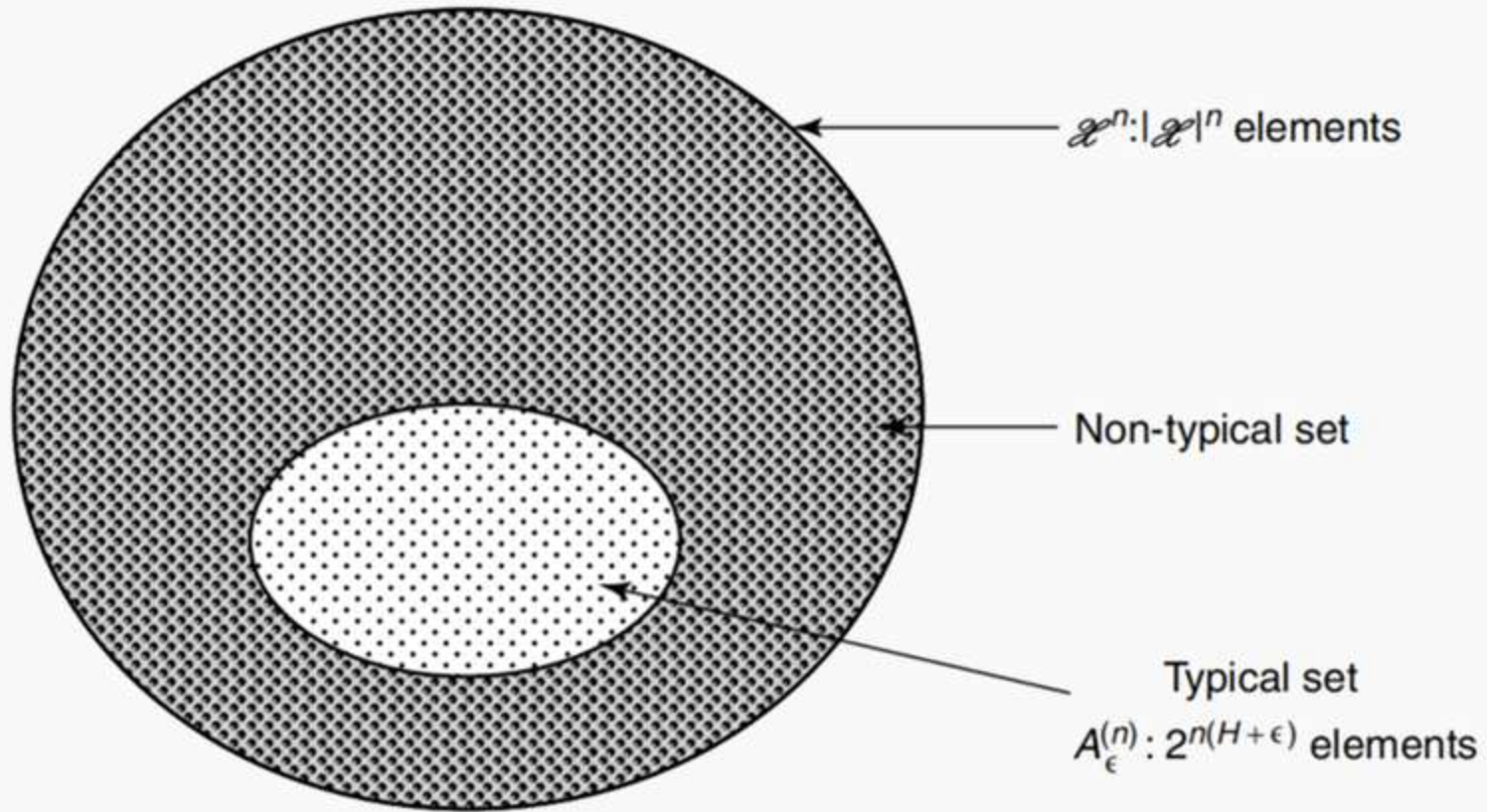
判断  $\mathcal{X}^n$  中哪些序列为典型序列。

- 解: 首先计算

$$H(X) = -\frac{25}{32} \log \frac{25}{32} - 7 \times \frac{1}{32} \log \frac{1}{32} \approx 1.39$$

对于  $\mathcal{X}^n$  中任一序列  $(x_1, x_2, \dots, x_n)$ , 如果其概率约等于  $2^{-1.39n}$ , 则为典型序列。典型序列的个数为  $2^{1.39n} \approx 2.62^n \ll 8^n$

# What is Typical Set?



# Fixed-length coding: **insights**

$nH(X)$  bits suffice on the average to describe  $n$  independent and *i.i.d.* r.v.'s.



$H(X)$  bits  $\rightarrow$  r.v.  $X$  ?

What is the **limitation** for fixed-length coding?

# Summary

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- Law of large numbers
- Asymptotic Equipartition Property (AEP)
- Typical Set
- Insights of the AEP

**AEP is central in information theory.**



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## 重难点:

- AEP定理及证明
- AEP的物理意义

# Thank you!

**Yayu Gao**

**School of Electronic Information and Communications  
Huazhong University of Science and Technology**

**Email: [yayugao@hust.edu.cn](mailto:yayugao@hust.edu.cn)**

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